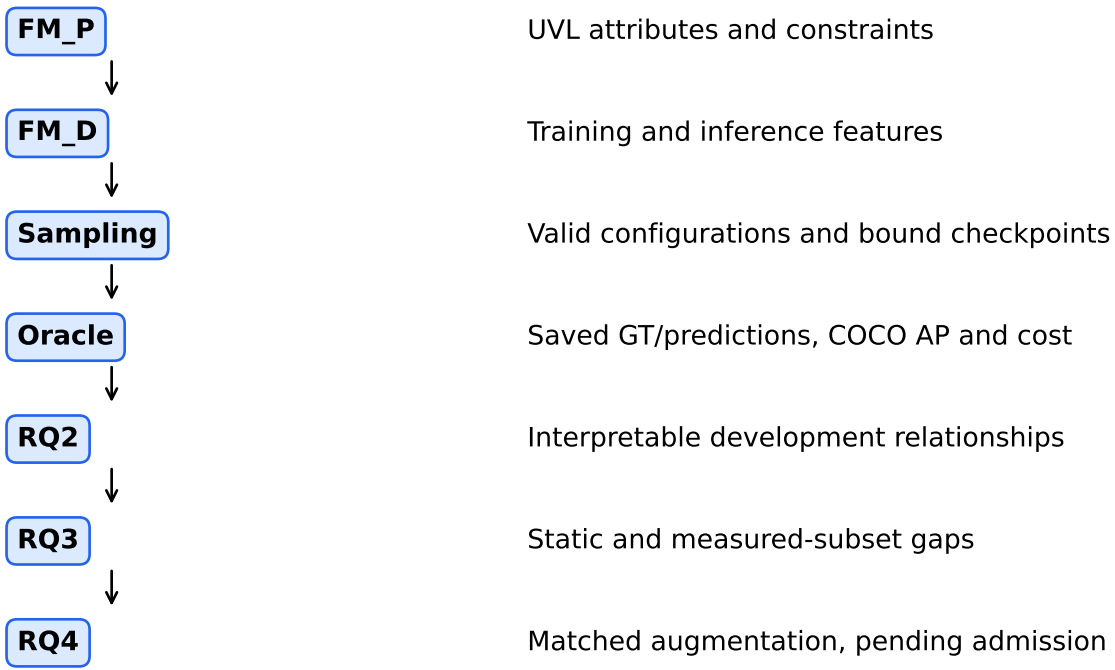
Dual variability research workflow



CONCEPTUAL_FRAMEWORK; completion tracked separately | Units: workflow steps

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: rq1_validation.json SHA256 dd7a8dcf4c31; full provenance in figure_evidence.json

Original UVL feature trees

FM_P

```
Degradation
    mandatory
        Illumination
            alternative
                Normal
                Overexposed_mild
                Overexposed_strong
                Underexposed_mild
                Underexposed_strong
                LowLight
    optional
        Blur
            or
                MotionBlur
                    alternative
                        Motion_k7
                        Motion_k15
                        Motion_k21
                DefocusBlur
                    alternative
                        Defocus_r3
                        Defocus_r6
        Noise
            or
                GaussianNoise
                    alternative
                        Gauss_s5
                        Gauss_s15
                        Gauss_s30
                ShotNoise
                    alternative
                        Shot_low
                        Shot_high
        Compression
            alternative
                JPEG_q75
                JPEG_q50
                JPEG_q25
        Atmosphere
            alternative
                Haze_light
                Haze_dense
                Rain
        Altitude
            alternative
                Downscale_x1_5
                Downscale_x2
        RollingShutter
```

FM_D

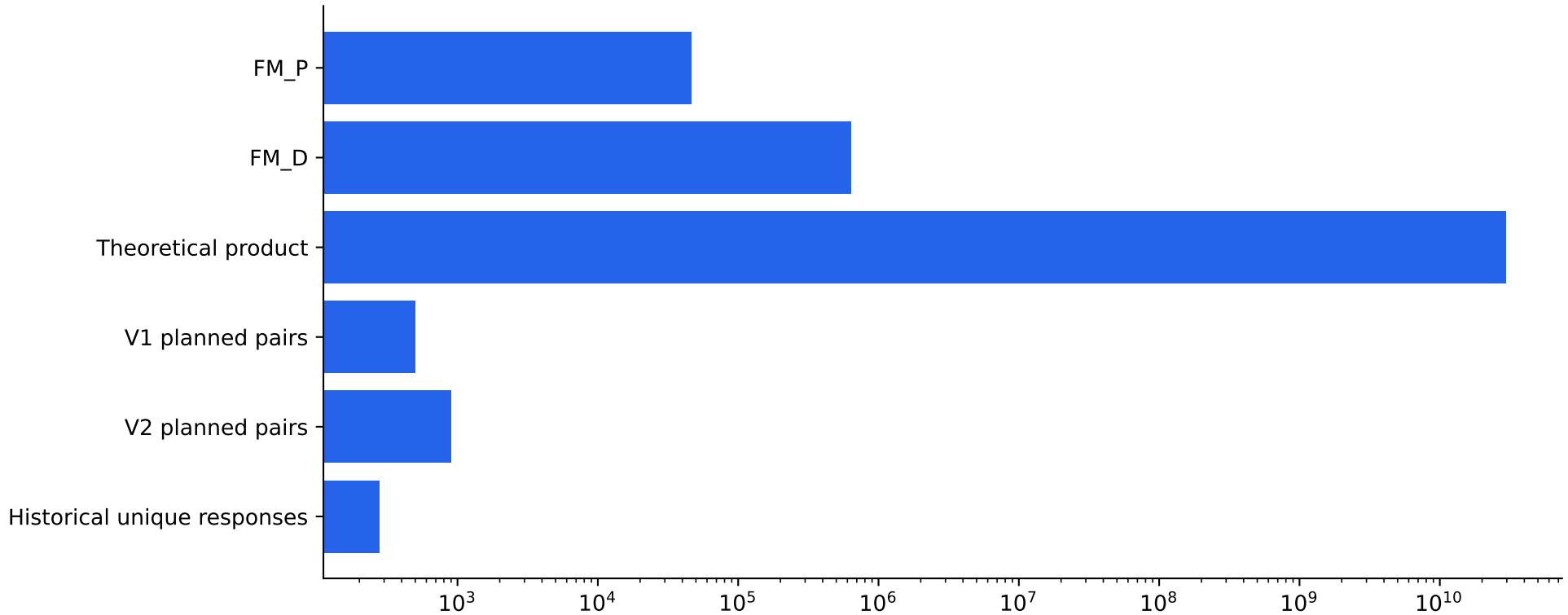
```
Detector
    mandatory
        Architecture
            alternative
                YOLOv8
                YOLO11
                YOLO26
                LEAF_YOLO
        ModelScale
            alternative
                Nano
                Small
        Weights
            alternative
                COCO_ZeroShot
                VisDrone_FT
        TrainResolution
            alternative
                Train640
                Train1280
        TrainAugmentation
            mandatory
                HSV_Default
                Geometric_Default
            optional
                Albu_Default
                NoiseAug
                MotionBlurAug
                ExposureAug
                JPEGAug
                HazeAug
        InferResolution
            alternative
                Infer640
                Infer960
                Infer1280
                Infer1536
        InferenceStrategy
            alternative
                FullFrame
                SAHI_Slicing
        PostProcessing
            alternative
                NMS
                EndToEnd_NMSFree
        Precision
            alternative
                FP32
                FP16
                INT8
    optional
        Preprocessing
            or
                CLAHE_Pre
                Gamma_Pre
                Denoise_Pre
                Deblur_Pre
        TTA
```

VALIDATED_LOGIC; not universal implementation support | Units: 25 / 34 leaves

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: fm_p.uvl SHA256 2033bb61456d; full provenance in figure_evidence.json

Configuration space and observed scope

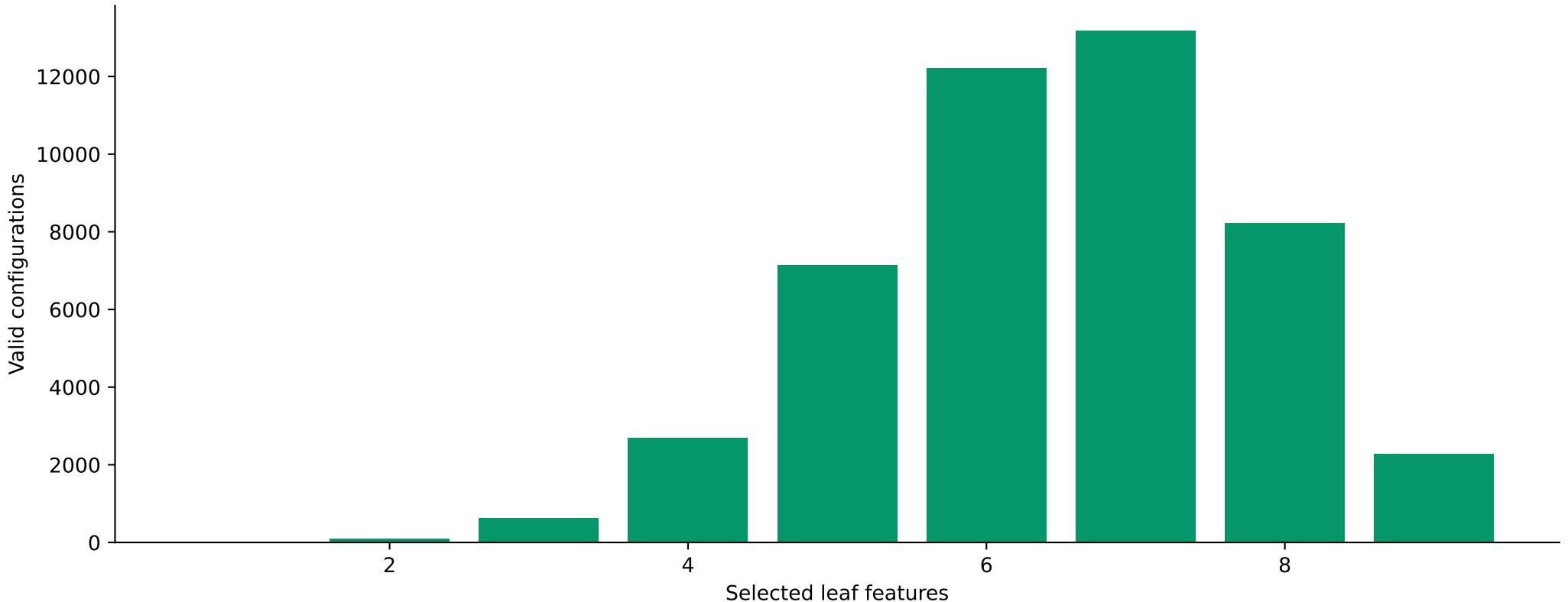


LOGICAL_COUNTS_AND_PLANS; historical responses are development only | Units: configuration count, logarithmic axis

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: rq1_validation.json SHA256 dd7a8dcf4c31; full provenance in figure_evidence.json

Exact distribution of valid FM_P configurations

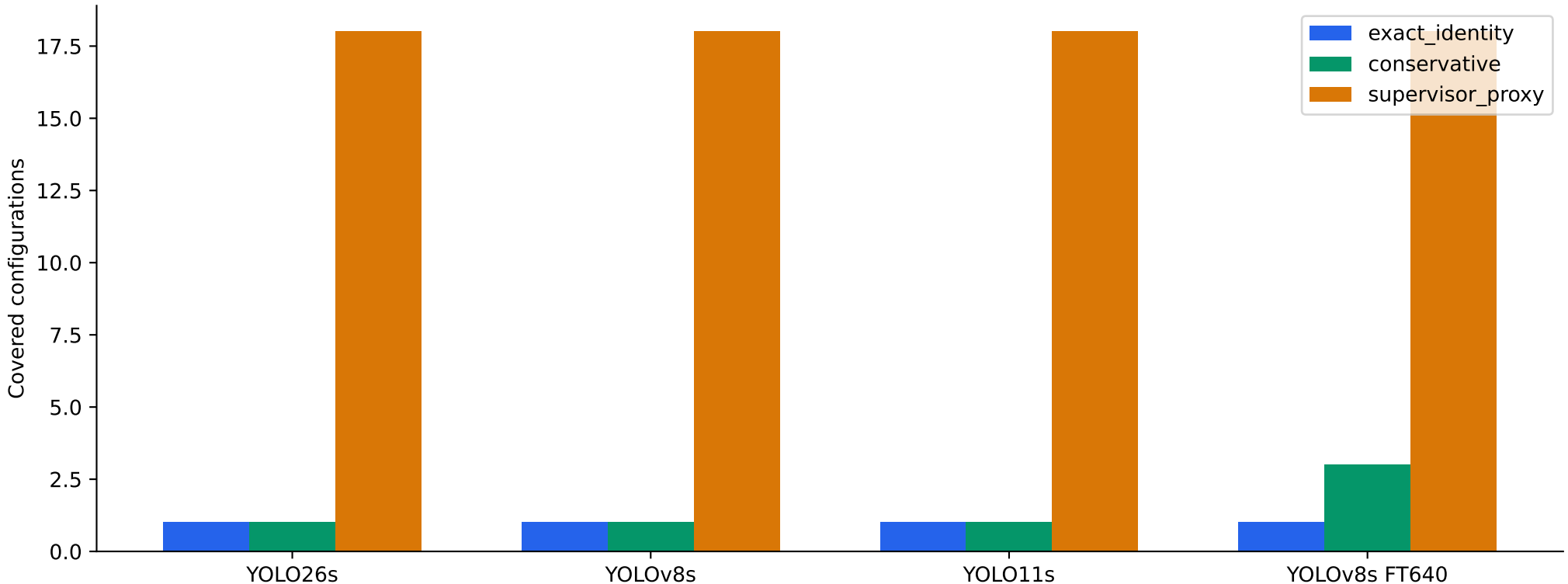


EXACT_ENUMERATION | Units: configuration count

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: configuration_distribution.csv SHA256 4d0b4b96742f; full provenance in figure_evidence.json

Static coverage depends on the evidence policy

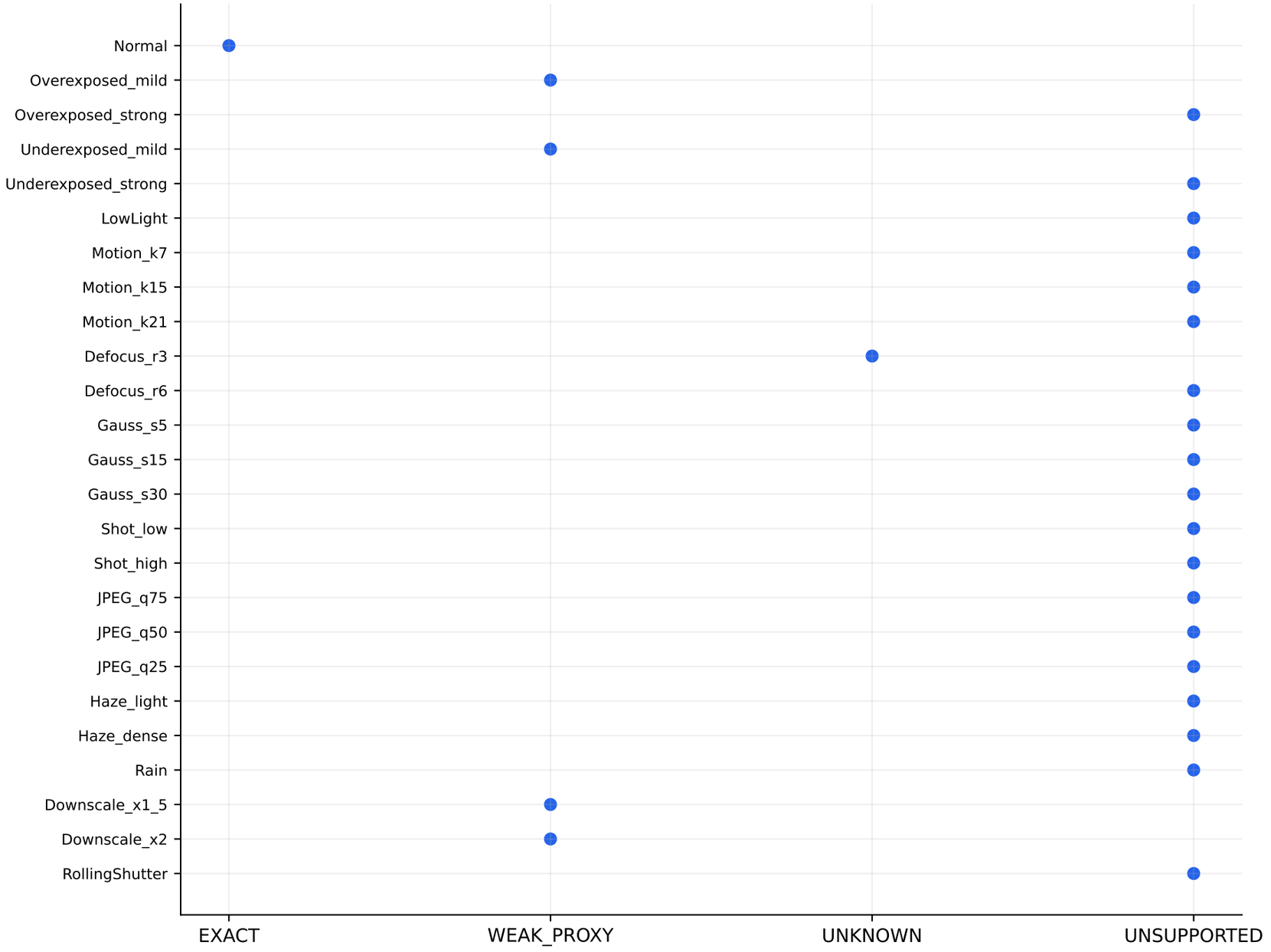


STATIC_ONLY; not detector failure | Units: covered FM_P configurations out of 46,368

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: static_coverage_by_checkpoint.csv SHA256 ef07d61fe9d3; full provenance in figure_evidence.json

Training-source coverage by degradation feature

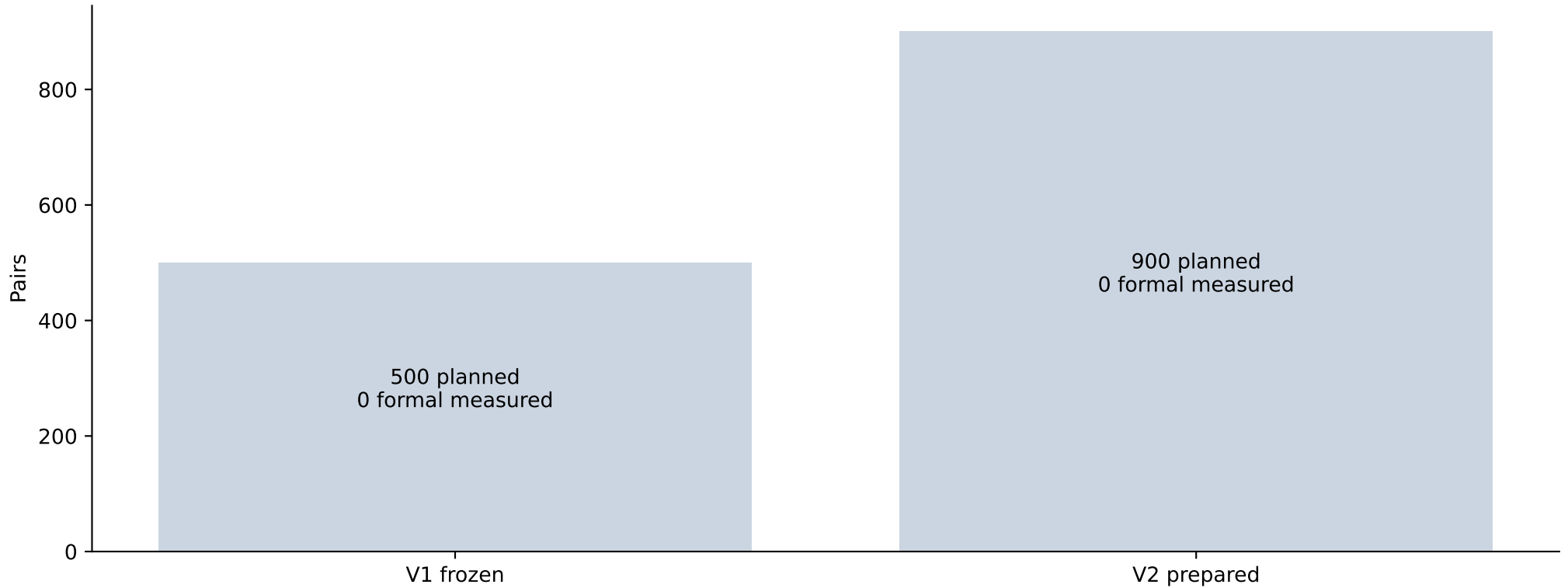


CV261102_SOURCE_POLICY; historical activation limits retained | Units: classification, not AP

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: static_baseline.json SHA256 b5cbacc3b975; full provenance in figure_evidence.json

Sampling design and formal execution coverage

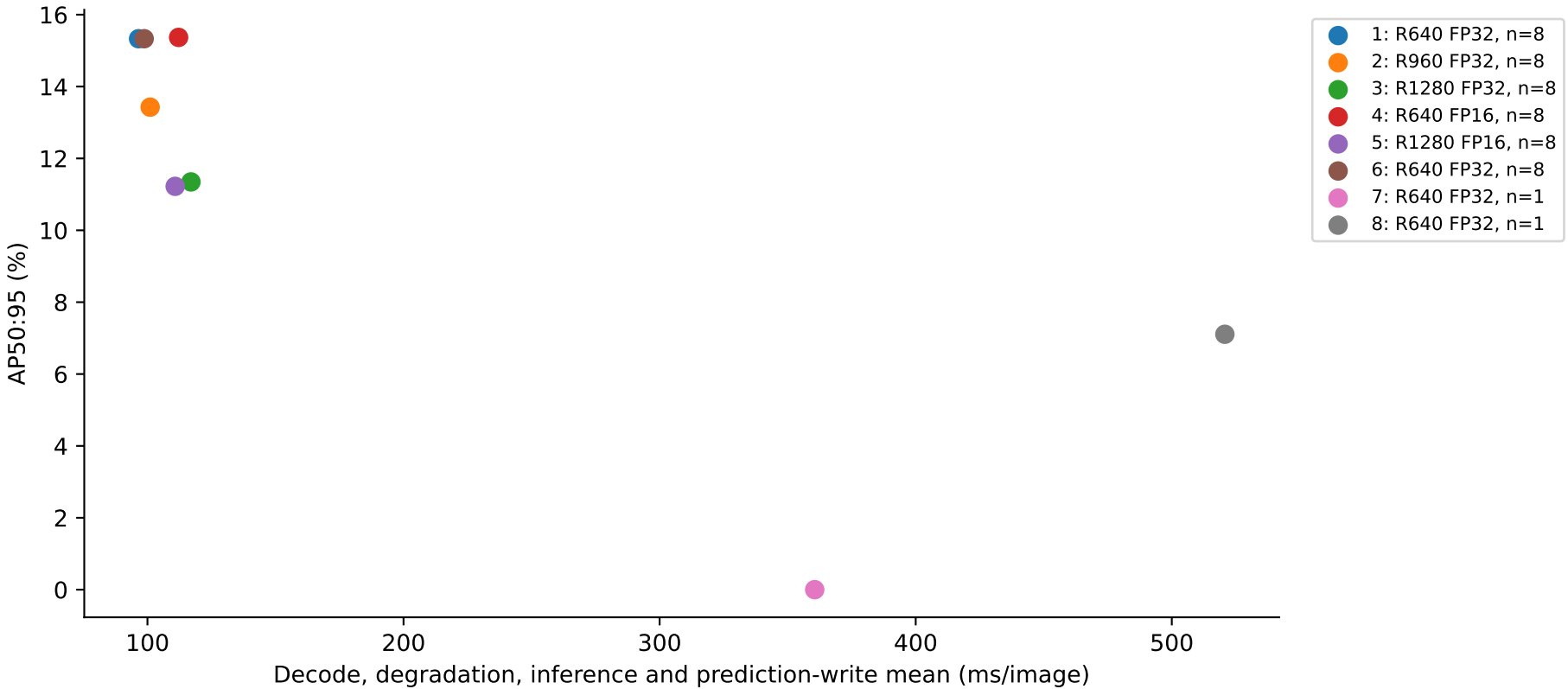


PLANNED_ONLY; DATASET_BLOCKED; no formal pairs | Units: pair count

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: cross_architecture_v2.json SHA256 e06b7a40d0d6; full provenance in figure_evidence.json

Saved engineering AP versus end-to-end image cost

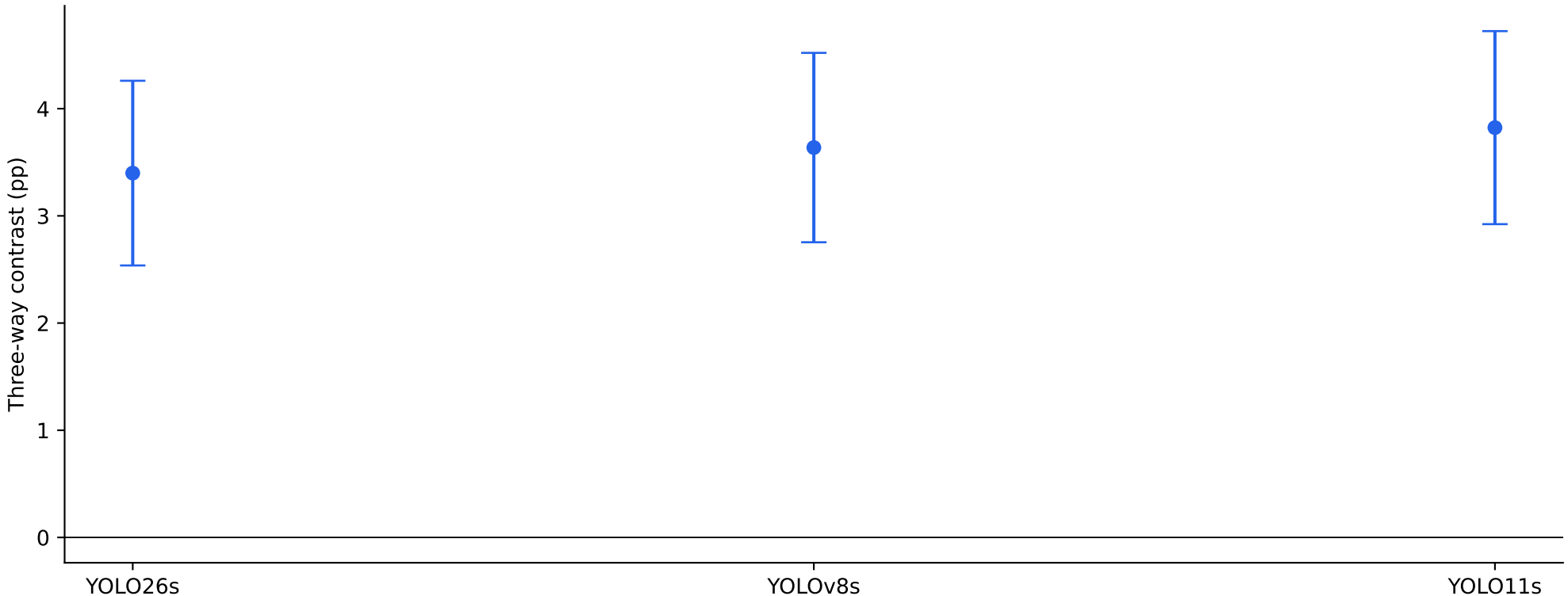


VAL_ENGINEERING_ONLY; image counts differ; not a matched accuracy comparison | Units: AP50:95 (%), ms/image

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: engineering_pairs.csv SHA256 7873a00a8934; full provenance in figure_evidence.json

Resolution × Blur × JPEG development interaction

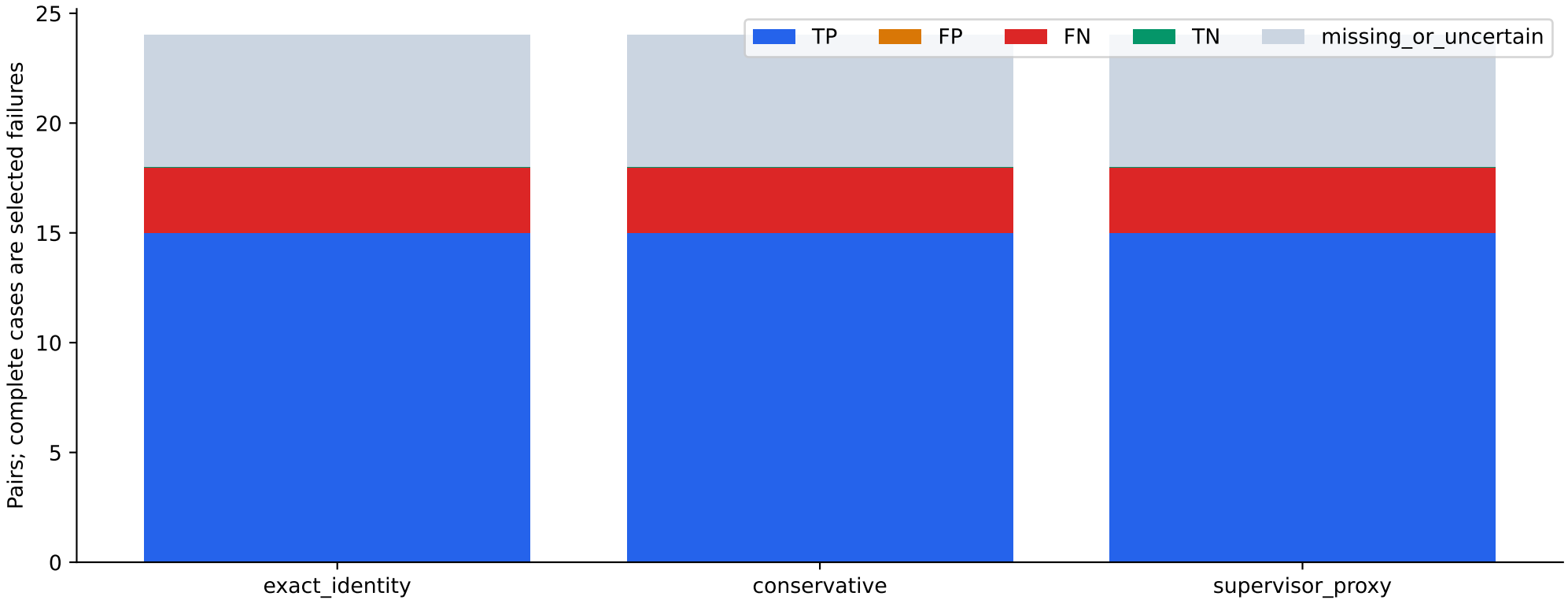


DEVELOPMENT_ONLY; 200 paired image draws, original joint36 family | Units: AP50:95 contrast (percentage points)

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63 | vm26_r001

Source: bootstrap_joint36.csv SHA256 fba3ad333d39; full provenance in figure_evidence.json

Static versus empirical: incomplete outcome diagnostics

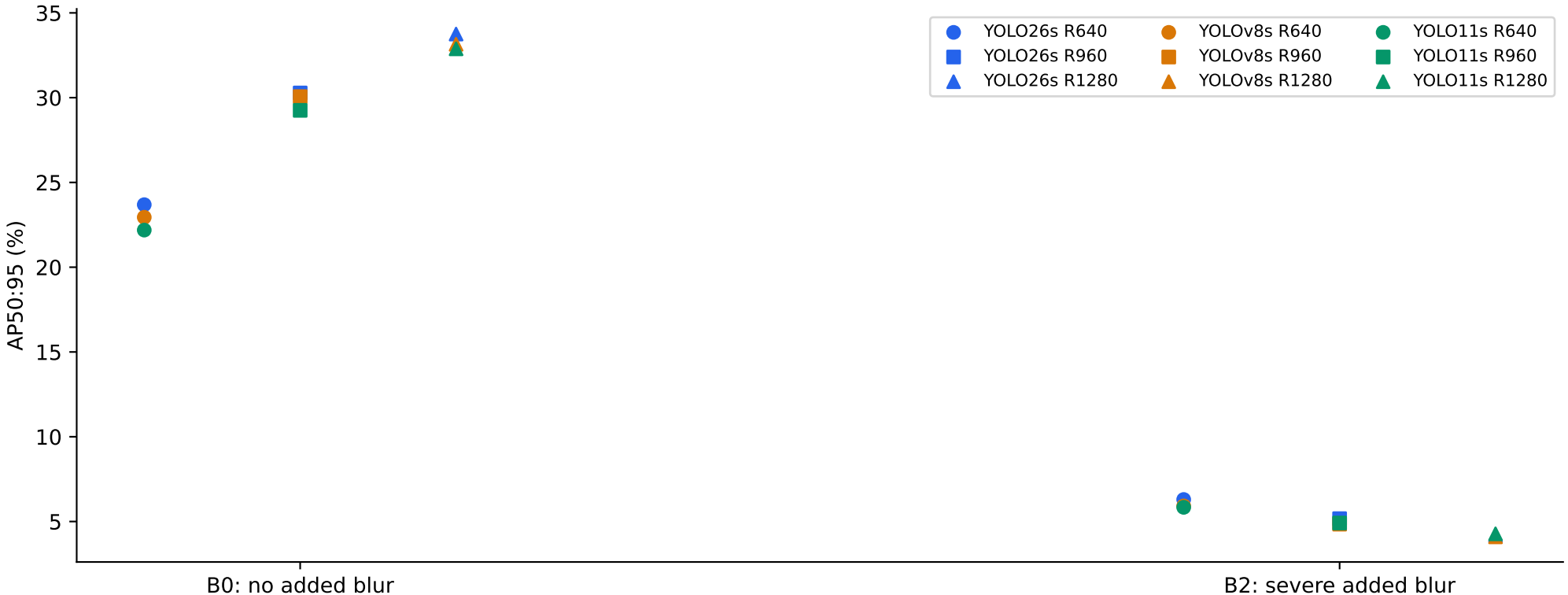


DEVELOPMENT_ONLY; compatible end-to-end latency is missing | Units: pair counts; tiny AP50:95 threshold 0.10

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: static_empirical_diagnostics.csv SHA256 4a8943cd538e; full provenance in figure_evidence.json

Development robustness across the two measured blur levels



DEVELOPMENT_ONLY; I0/N0; mean over three measured extra-JPEG settings | Units: AP50:95 (%); no intermediate blur inferred

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63 | vm26_r001

Source: configuration_metrics.csv SHA256 ffbff5d3476d; full provenance in figure_evidence.json

Targeted augmentation and gap closure

NOT_EVALUATED

Matched protocol prepared. Training executor and health canary pending.
Representative baseline gap, independent admission and measured budget remain required.

NOT_EVALUATED; no matched training or before/after response | Units: no measured gain

CV261207 cv261207_rq_figures_20261107T221015_cf1dcb63

Source: rq4_matched_protocol.json SHA256 40671c446648; full provenance in figure_evidence.json